Doiraviy diagrammani o'qish va talqin qilish

Doiraviy diagrammadagi son, kasr va foizlardan har bo'lakda nechta narsa borligini va jami qancha ekanini topamiz.

1-QISM

Mavzu

ESLAB OLAYLIK

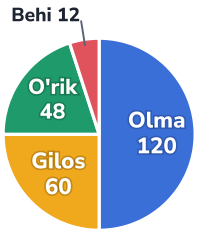
- Butun doira — 1 butun, 100%. Yarim doira (diametr bilan ajratilgan) — 50%, to'g'ri burchak belgili bo'lak — chorak, 25%.
- 72 ning $\frac{5}{8}$ qismi — $72 : 8 \cdot 5 = 45$; 600 ning 15% i — $600 : 100 \cdot 15 = 90$.

USUL · BO'LAKLARDA SONLAR YOZILGAN

- Qanchaga ko'p — ayiramiz; necha marta ko'p — bo'lamiz.
- Jami — hamma bo'laklardagi sonlarni qo'shamiz.
- Qancha qismi — bo'lakdagi son : jami (kasr); necha foizi — shu kasr $\cdot 100\%$.

1-MISOL · FARQ, JAMI, QISM

Bog'dagi mevali daraxtlar.
a) Olma gilosdan nechta ko'p? b) Jami nechta daraxt bor?
c) Daraxtlarning qancha qismi o'rik?



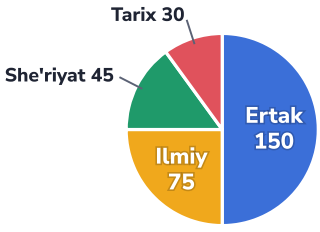
a) $120 - 60 = 60$ ta.

b) $120 + 60 + 48 + 12 = 240$ ta.

c) $\frac{48}{240} = \frac{1}{5}$ qismi ($48 \cdot 5 = 240$).

2-MISOL · FOIZ VA NECHA MARTA

Kutubxonaga kelgan yangi kitoblar.
a) Jami nechta kitob keldi? b) Necha foizi ilmiy?
c) Ertak kitoblar tarix kitoblaridan necha marta ko'p?



a) $150 + 75 + 45 + 30 = 300$ ta.

b) $\frac{75}{300} = \frac{1}{4} = 25\%$.

c) $150 : 30 = 5$ marta.

USUL · BO'LAKLARDA KASR YOKI FOIZ YOZILGAN, JAMI MA'LUM

- Eng katta bo'lak — eng ko'p tanlangan narsa.
- Bo'lakdagi son = kasr · jami. Foiz bo'lsa: jami : 100 · foiz.
- Yozilmagan bo'lak: 1 dan (100% dan) qolgan bo'laklarni ayiramiz.

3-MISOL · KASRLAR

60 o'quvchining sevimli fasli.
a) Qaysi fasl eng ko'p yoqadi?
b) O'quvchilarning qancha qismi yozni yaxshi ko'radi?
c) Kuzni necha o'quvchi yaxshi ko'radi?
d) Bahorni yozdan ko'ra nechta ko'p o'quvchi tanlagan?



a) Eng katta bo'lak — bahor.

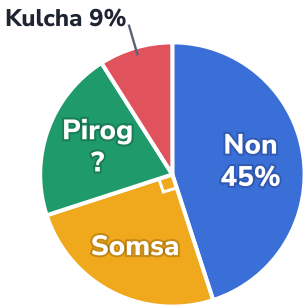
b) $1-\frac{2}{5}-\frac{1}{4}-\frac{1}{6}=\frac{60-24-15-10}{60}=\frac{11}{60}$.

c) $\frac{1}{6}\cdot 60=10$ nafar.

d) $\frac{24}{60}-\frac{11}{60}=\frac{13}{60}$; $\frac{13}{60}\cdot 60=13$ nafar.

4-MISOL · FOIZLAR

Novvoyxona bir kunda 600 ta mahsulot pishirdi.
a) Necha foizi somsa? b) Necha foizi pirog?
c) Nechta non pishirildi?
d) Non kulchadan necha marta ko'p?



a) To'g'ri burchak belgisi — chorak: 25%.

b) $100\%-45\%-25\%-9\%=21\%$.

c) $600:100\cdot 45=270$ ta.

d) $45\%:9\%=5$ marta ($270:54=5$).



Somsa bo'lagiga son yozilmagan. Lekin to'g'ri burchak belgisi bor — demak, u doiraning to'rtdan biri, ya'ni 25%.

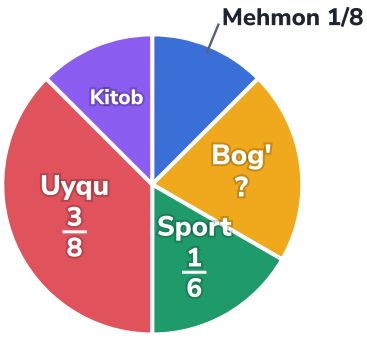
5-MISOL · YARIM DOIRA

Diagrammada Sardorning dam olish kunidagi 24 soati ko'rsatilgan. O'ng yarim doirada uchta ish bor: mehmonga borish, bog'da ishlash va sport.

a) Uyquga necha soat ketgan?

b) Bog'da ishlash sutkaning qancha qismini olgan?

c) Kitob o'qishga ketgan vaqtning butun sutkaga nisbati qanday?



a $\frac{3}{8} \cdot 24 = 9$ soat.

b O'ng yarim doira — $\frac{1}{2}$. Undan qolgan ikki ishni ayiramiz: $\frac{1}{2} - \frac{1}{8} - \frac{1}{6} = \frac{12}{24} - \frac{3}{24} - \frac{4}{24} = \frac{5}{24}$.

c Chap yarim doirada uyqu va kitob bor. Kitob: $\frac{1}{2} - \frac{3}{8} = \frac{4}{8} - \frac{3}{8} = \frac{1}{8}$. Nisbat — $1 : 8$ (3 soat va 24 soat).

USUL · BITTA BO'LAKNING SONI MA'LUM — JAMINI TOPAMIZ

- Bo'lak doiraning $\frac{1}{4}$ qismi bo'lsa: jami = son · 4 (yarim doira bo'lsa: · 2).
- Kasri yoki foizi berilgan bo'lsa: jami = son : kasr; foiz bo'lsa — son : foiz · 100.
- «2 marta ko'p» bo'lsa: qolgan qism 2 + 1 = 3 ulushga bo'linadi.

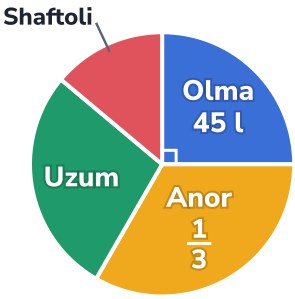
6-MISOL · CHORAK BO'LAK

Do'kon bir kunda to'rt xil sharbat sotdi. Uzum sharbati shaftolinikidan 2 marta ko'p.

a) Jami necha litr sharbat sotildi?

b) Anor sharbati necha litr?

c) Shaftoli sharbati necha litr?



a) Olma — chorak bo'lak. Jami: $45 \cdot 4 = 180$ l.

b) $\frac{1}{3} \cdot 180 = 60$ l.

c) Uzum va shaftoli birga: $180 - 45 - 60 = 75$ l. Bu 3 ulush (uzum — 2, shaftoli — 1). Shaftoli: $75 : 3 = 25$ l, uzum — 50 l.

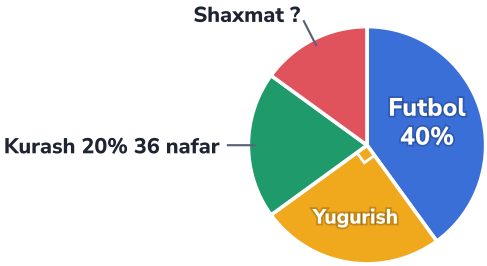
7-MISOL · FOIZI MA'LUM BO'LAK

Sport bayramida to'rt xil musobaqa bo'ldi. Kurashda 36 o'quvchi qatnashdi.

a) Shaxmatda necha foiz o'quvchi qatnashdi?

b) Jami necha o'quvchi qatnashdi?

c) Futbolda nechta o'quvchi qatnashdi?



a) Yugurish — chorak, 25%. $100\% - 40\% - 25\% - 20\% = 15\%$.

b) 20% — 36 nafar, 10% — 18 nafar. Jami $18 \cdot 10 = 180$ nafar.

c) $180 : 100 \cdot 40 = 72$ nafar.

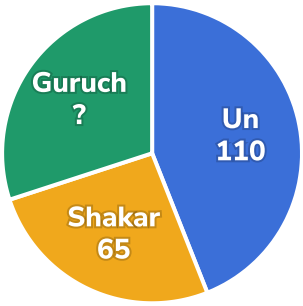
2-QISM

Misollar ishlash

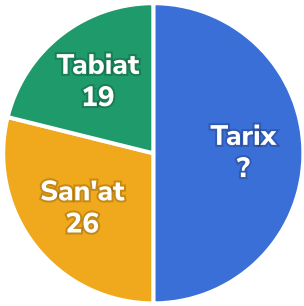
1-MISOL · BITTA SON YOZILMAGAN

a) Omborda 250 qop mahsulot bor. Guruch necha qop?

b) Sayyohlar uchta muzeydan birini tanladi. Tarix muzeyini necha sayyoh tanladi?



a) 250 qop



b) Muzeylar

a Jami ma'lum. Ma'lum bo'laklarni ayiramiz: $250 - 110 - 65 = 75$ qop.

b Jami berilmagan, lekin Tarix bo'lagi — yarim doira. Qolgan yarmi: $26 + 19 = 45$. Demak, Tarix ham 45 kishi.

Shart o'zgarsa: b) da jami so'ralsa — $45 \cdot 2 = 90$ kishi. Bo'lak chorak bo'lsa, qolgan bo'laklar birga $\frac{3}{4}$ bo'ladi: ularning yig'indisini 3 ga bo'lsak, chorak bo'lakdagi son chiqadi.

2-MISOL · «... MARTA KO'P» SHARTI

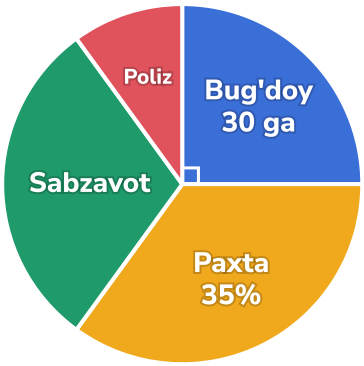
Fermerning yeri to'rt ekinga bo'lingan. Bug'doy 30 gektar. Sabzavot polizdan 3 marta ko'p.

a) Sabzavot va poliz birga yerning necha foizi?

b) Poliz necha foiz?

c) Fermerning jami yeri necha gektar?

d) Sabzavot necha gektar?



a Bug'doy — chorak, 25%. $100\% - 25\% - 35\% = 40\%$.

b Sabzavot — 3 ulush, poliz — 1 ulush, jami 4 ulush. Poliz: $40\% : 4 = 10\%$, sabzavot — 30%.

c $30 \cdot 4 = 120$ gektar.

d $120 : 100 \cdot 30 = 36$ gektar.

3-MISOL · NISBAT

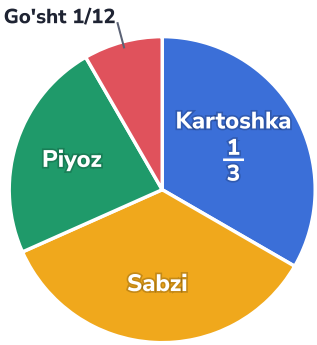
Maktab oshxonasi bir haftada 240 kg mahsulot ishlatdi. Sabzi va piyoz nisbati 3 : 2.

a) Sabzi va piyoz birga qancha qismni tashkil qiladi?

b) Sabzi necha kg?

c) Go'shtning jami mahsulotga nisbati qanday?

d) Kartoshkaning go'shtga nisbati qanday?



a $1 - \frac{1}{3} - \frac{1}{12} = \frac{12}{12} - \frac{4}{12} - \frac{1}{12} = \frac{7}{12}$.

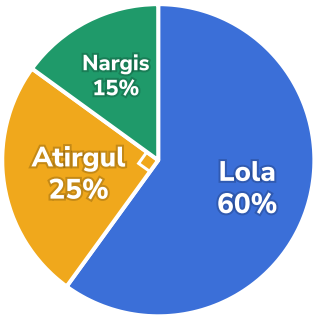
b $\frac{7}{12} \cdot 240 = 140$ kg. $3 + 2 = 5$ ulush, bir ulush $140 : 5 = 28$ kg. Sabzi: $28 \cdot 3 = 84$ kg.

c Go'sht — $\frac{1}{12}$ qism, nisbat $1 : 12$.

d $\frac{1}{3} = \frac{4}{12}$, go'sht $\frac{1}{12}$. Nisbat $4 : 1$ (80 kg va 20 kg).

4-MISOL · MA'LUMOT O'ZGARSA

Gulzorda 24 ta lola, 10 ta atirgul va 6 ta nargis bor edi. Diagrammada shu holat ko'rsatilgan. Keyin yana 20 ta atirgul ekildi. Yangi diagrammada har bir gul necha foiz bo'ladi?



Oldin — 40 ta gul

1. Yangi jami: $24 + 30 + 6 = 60$ ta (40 ta emas).
2. Lola: $\frac{24}{60} \cdot 100\% = 40\%$; atirgul: $\frac{30}{60} \cdot 100\% = 50\%$; nargis: $\frac{6}{60} \cdot 100\% = 10\%$.
3. Tekshiramiz: $40\% + 50\% + 10\% = 100\%$. Lolalar soni o'zgarmadi, lekin foizi 60% dan 40% ga tushdi.

Shart o'zgarsa: «qaysi diagramma to'g'ri?» deb so'ralsa — yangi foizlarni topib, diagrammalardagi bilan solishtiramiz. Ikkinchi gulzorga oldingi diagrammadagi foizlar bilan 200 ta gul ekilsa: lola $200 : 100 \cdot 60 = 120$ ta, atirgul 50 ta, nargis 30 ta. Bunda foizlar o'zgarmaydi.

5-MISOL · KASRI MA'LUM BO'LAK

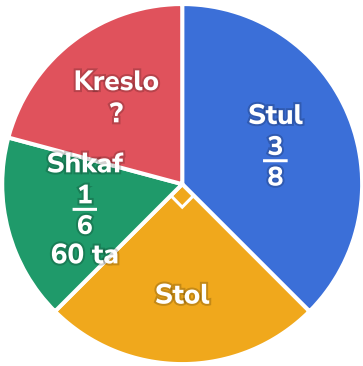
Mebel sexi bir oyda to'rt xil buyum yasadi. Shkaflar 60 ta.

a) Buyumlarning qancha qismi kreslo?

b) Sex jami nechta buyum yasadi?

c) Nechta kreslo yasaldi?

d) «Stullar kreslolardan 2 marta ko'p». Bu fikr to'g'rimi?



a Stol — chorak, $\frac{1}{4}$. $1 - \frac{3}{8} - \frac{1}{4} - \frac{1}{6} = \frac{24}{24} - \frac{9}{24} - \frac{6}{24} - \frac{4}{24} = \frac{5}{24}$.

b $\frac{1}{6}$ qism — 60 ta. Jami: $60 : \frac{1}{6} = 60 \cdot 6 = 360$ ta.

c $360 : 24 \cdot 5 = 75$ ta.

d Stullar: $360 : 8 \cdot 3 = 135$ ta. Kreslolardan 2 marta ko'p bo'lsa, $75 \cdot 2 = 150$ ta bo'lardi. Fikr noto'g'ri.

Shart o'zgarsa: bo'lakka foiz yozilgan bo'lsa, masalan «15% — 45 ta», jami $45 : 15 \cdot 100 = 300$ ta bo'ladi.

Shkaflar — doiraning oltidan bir bo'lagi. Butun doirada shunday 6 ta bo'lak bor, shuning uchun 60 ni 6 ga ko'paytirdik.



6-MISOL · JADVALDAN DIAGRAMMAGA

6-sinf o'quvchilari maktab kutubxonasiga a'zo yoki a'zo emas.

a) Jadvaldagi bo'sh kataklarni to'ldiring.

b) Qaysi diagramma a'zo va a'zo bo'lmaganlarni to'g'ri ko'rsatadi?

	A'zo	A'zo emas	Jami
O'g'il bolalar	31	14	45
Qizlar	29	16	45
Jami	60	30	90



A



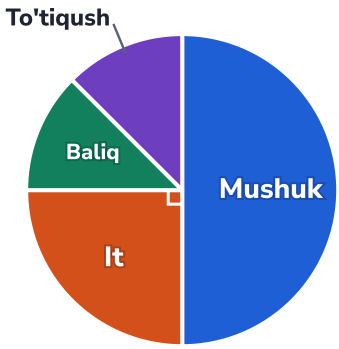
B

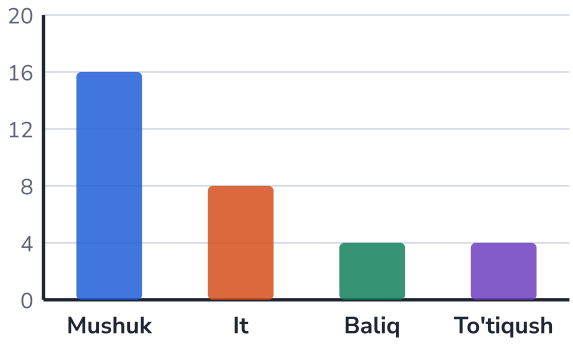
a O'g'il a'zo: $45 - 14 = 31$. Qizlar jami: $90 - 45 = 45$. Qiz a'zo emas: $45 - 29 = 16$. A'zo: $31 + 29 = 60$, a'zo emas: $14 + 16 = 30$.

b A'zolar $\frac{60}{90} = \frac{2}{3}$, a'zo emaslar $\frac{1}{3}$. B da a'zo emaslar chorak ($\frac{1}{4}$) — to'g'ri emas. Javob: A.

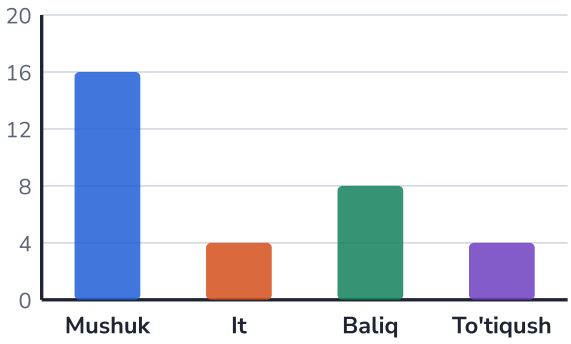
7-MISOL · MOS USTUNLI DIAGRAMMA

Diagrammada 32 o'quvchining sevimli uy hayvoni ko'rsatilgan. Baliq va to'tiqush bo'laklari teng. Qaysi ustunli diagramma shu ma'lumotni to'g'ri ko'rsatadi?





A



B

1. Mushuk — yarim doira: $32 : 2 = 16$. It — chorak: $32 : 4 = 8$.
2. Baliq va to'tiqush teng, qolgan $32 - 16 - 8 = 8$ tani bo'lishadi: 4 tadan.
3. A da 16, 8, 4, 4 — mos. B da it va baliq almashib qolgan. Javob: A.

8-MISOL · QAYSI DIAGRAMMA KERAK?

Qaysi ma'lumotni doiraviy diagrammada ko'rsatgan ma'qul?

a) Bir hafta davomida har kuni sotilgan non soni.

b) Oila oylik daromadining nimalarga sarflangani.

b. Butun daromad bo'laklarga bo'linadi — har bo'lak butunning qismi. Bunga doiraviy diagramma mos.

a. Bu yerda butun yo'q, son kundan kunga o'zgaradi. Uni ustunli yoki chiziqli diagrammada ko'rsatgan ma'qul.



Gulzorga atirgul qo'shilganda lolalar soni o'zgarmadi, ammo foizi kamaydi. Foizni har doim yangi jamidan hisoblash kerak.

DIQQAT!

1. Foizni son deb o'qimang. 400 kishining 25% i: ✗ 25 kishi ✓ $400 : 100 \cdot 25 = 100$ **kishi**
2. Bo'lakning soni va kasri ma'lum bo'lsa, jamini bo'lish bilan topamiz: ✗ $60 \cdot \frac{1}{6}$ ✓ $60 : \frac{1}{6} = 60 \cdot 6 = 360$
3. Yarim doiradagi bo'laklardan birini topishda 1 dan emas, $\frac{1}{2}$ dan ayiramiz: ✗ $1 - \frac{1}{8} - \frac{1}{6}$ ✓ $\frac{1}{2} - \frac{1}{8} - \frac{1}{6} = \frac{5}{24}$
4. «3 marta ko'p» bo'lsa, qolgan qism 3 ga emas, $3 + 1 = 4$ ulushga bo'linadi.
5. Javobni tekshiring: hamma bo'laklarning kasrlari yig'indisi 1 ga, foizlari yig'indisi 100% ga teng bo'lishi kerak.

3-QISM

Mustahkamlash

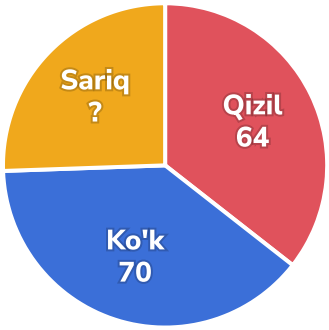
1. Diagrammadagi «?» o'rniga mos sonni toping.

   a) Do'konda 180 ta shar bor. Nechta shar sariq?

   b) Javondagi kitoblar. Nechta kitob darslik?

   c) Sayyohlar guruhi qaysi shaharga borishni tanlagan. Guruhning qancha qismi Buxoroni tanlagan?

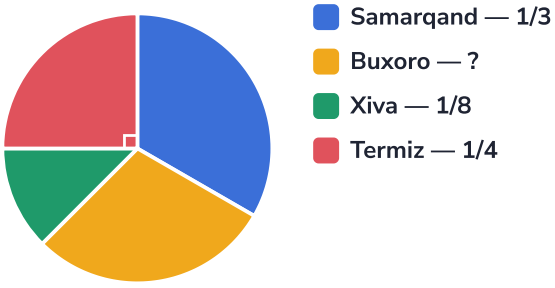
   d) Tumandagi ekin maydoni. Maydonning necha foizi poliz?



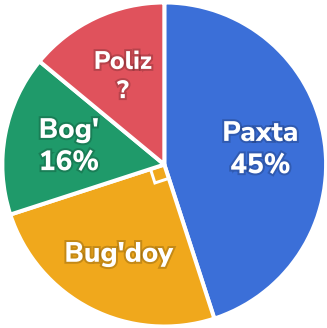
a)



b)

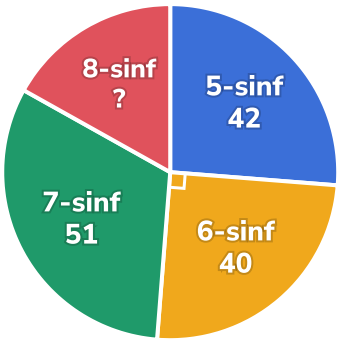


c)

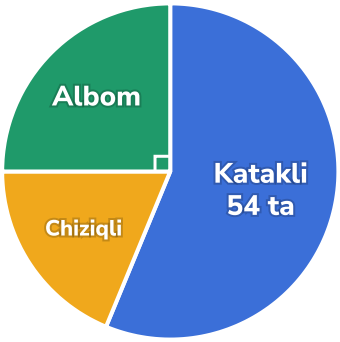


d)

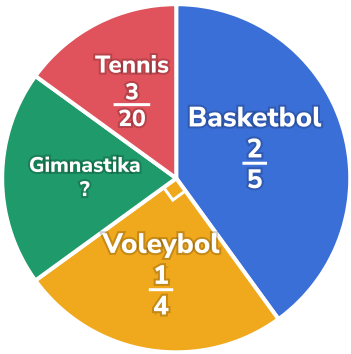
2. Diagrammada teatrga kelgan to'rt sinf o'quvchilari ko'rsatilgan.
   a) Teatrga jami nechta o'quvchi keldi?
   b) 8-sinfdan nechta o'quvchi keldi?



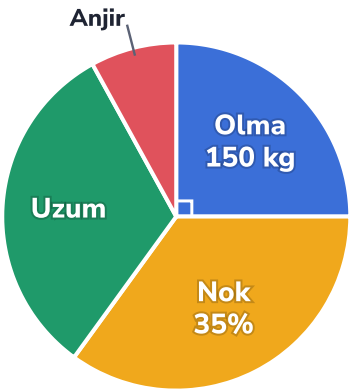
3. Kitob do'koni bir kunda uch xil daftar sotdi. Katakli daftarlar chiziqli daftarlardan 3 marta ko'p sotildi.
   a) Sotilgan daftarlarning qancha qismi albom?
   b) Katakli daftarlar qancha qismni tashkil qiladi?
   c) Jami nechta daftar sotildi?
   d) Chiziqli daftarlar nechta?



4. Diagrammada sport zalining bir haftalik mashg'ulot vaqti ko'rsatilgan. Tennisga 6 soat ajratilgan.
   a) Gimnastikaga vaqtning qancha qismi ajratilgan?
   b) Bir haftada jami necha soat mashg'ulot bo'ladi?
   c) Gimnastikaga necha soat ajratilgan?
   d) Basketbolga voleyboldan necha soat ko'p vaqt ajratilgan?



5. Meva rastasi bir kunda to'rt xil meva sotdi. Uzum anjirdan 4 marta ko'p sotildi.
   a) Uzum va anjir birga necha foizni tashkil qiladi?
   b) Anjir necha foiz?
   c) Jami necha kg meva sotildi?
   d) 1 kg anjir 25 000 so'm. Anjirdan qancha pul tushdi?

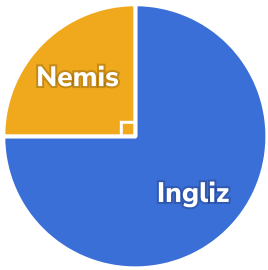


6. 6-sinf o'quvchilari ingliz yoki nemis tilini o'rganadi.
   a) Jadvaldagi bo'sh kataklarni to'ldiring.
   b) Qaysi diagramma ingliz va nemis tilini o'rganuvchilarni to'g'ri ko'rsatadi?

	Ingliz tili	Nemis tili	Jami
O'g'il bolalar	30	?	36
Qizlar	?	12	?
Jami	?	?	72

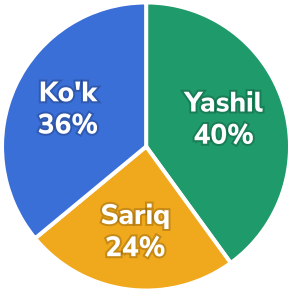


A

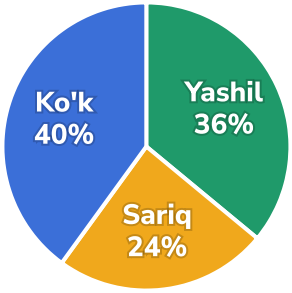


B

7. Qafasda 10 ta yashil, 6 ta sariq va 4 ta ko'k to'tiqush bor edi. Yana 5 ta ko'k to'tiqush olib kelindi. Qaysi diagramma yangi holatni to'g'ri ko'rsatadi? Tushuntiring.



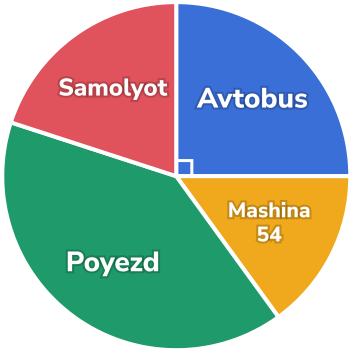
A



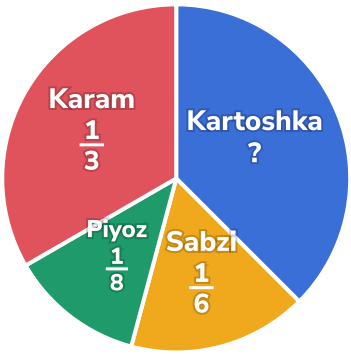
B

8. Bir maktab 40 ta daraxt ekdi: 20 ta olma, 12 ta o'rik va 8 ta gilos. Ikkinchi maktab 100 ta daraxt ekmoqchi. Uning doiraviy diagrammasi birinchi maktabnikidek bo'lishi kerak.
   a) Ikkinchi maktab har bir daraxtdan nechtadan ekishi kerak?
   b) Olma daraxtlarining foizi o'zgaradimi?

9. 360 nafar sayyoh shaharga turli transportda keldi. Poyezdda kelganlar samolyotda kelganlardan 2 marta ko'p. Necha sayyoh keldi:
   a) avtobusda; b) samolyotda; c) poyezdda?



10. Fermer to'rt xil sabzavot yig'di. Piyoz 6 tonna.
    a) Hosilning qancha qismi kartoshka?
    b) Jami necha tonna hosil yig'ildi? Kartoshka necha tonna?
    c) «Sabzi va karam birga kartoshkaga teng». Bu fikr to'g'rimi? Tushuntiring.

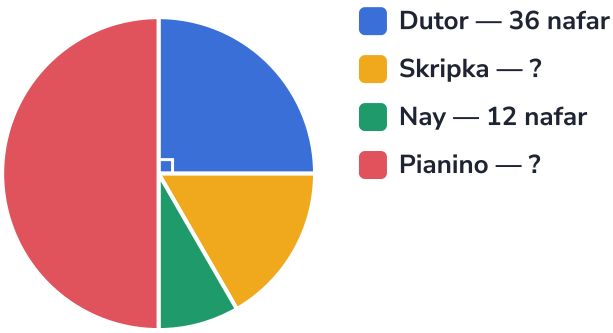


11. Qaysi ma'lumotni doiraviy diagrammada ko'rsatish qulay? Nima uchun?
    a) Bitta o'quvchining bo'yi 1-sinfdan 6-sinfgacha qanday o'sgani.
    b) Bir kunlik ovqatda oqsil, yog' va uglevod ulushi.
    c) Bir yil davomida har oyda yoqqan yomg'ir miqdori.
    d) Sinf sardori saylovida har bir nomzodga berilgan ovozlar.

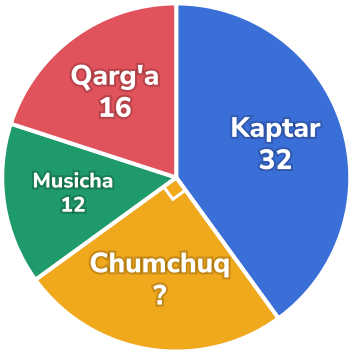
4-QISM

Mustaqil ish

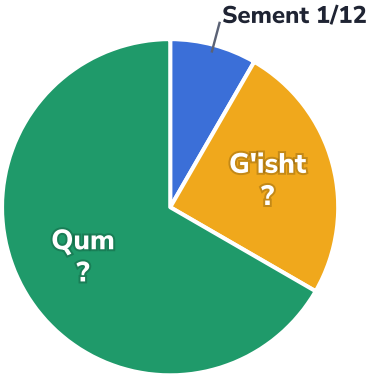
1. Musiqa maktabida to'rt xil cholg'u o'rgatiladi. Nechta o'quvchi o'rganadi:
   a) pianino chalishni; b) skripka chalishni?



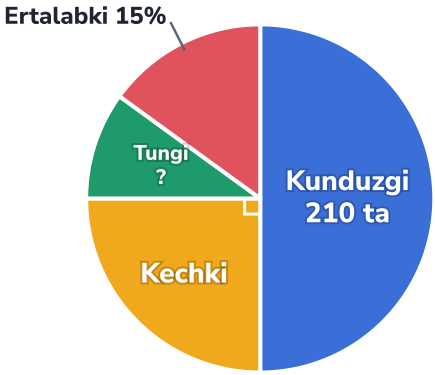
2. Bog'dagi qushlar sanaldi.
   a) Kaptarlar qarg'alardan necha marta ko'p?
   b) Chumchuqlar nechta?
   c) Qushlarning qancha qismi musicha?



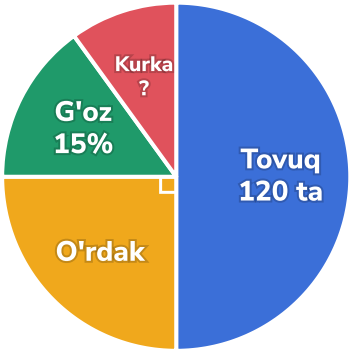
3. Qurilishga uch xil material keltirildi. G'isht sementdan 3 marta ko'p. Javobni qisqarmas kasr bilan yozing.
   a) Materiallarning qancha qismi g'isht?
   b) Qancha qismi qum?



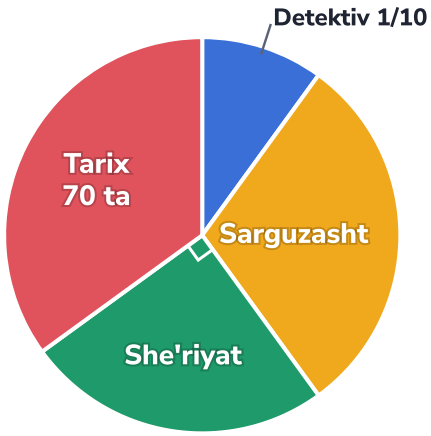
4. Kinoteatr bir kunda to'rt seansga chipta sotdi.
   a) Jami nechta chipta sotildi?
   b) Tungi seansga nechta chipta sotildi?



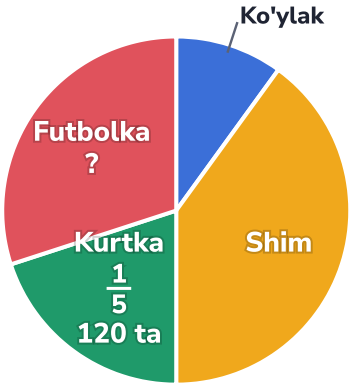
5. Diagrammada fermadagi parrandalar ko'rsatilgan.
   a) Parrandalarning necha foizi kurka?
   b) Tovuqlar kurkalardan necha marta ko'p?
   c) Fermada jami nechta parranda bor?
   d) G'ozlar nechta?



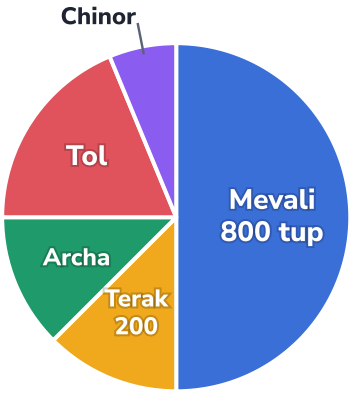
6. Kutubxonadagi kitoblar. Sarguzasht kitoblar detektivdan 3 marta ko'p.
   a) Kitoblarning qancha qismi sarguzasht?
   b) Qancha qismi tarix kitoblari?
   c) Kutubxonada jami nechta kitob bor?
   d) Sarguzasht kitoblar nechta?



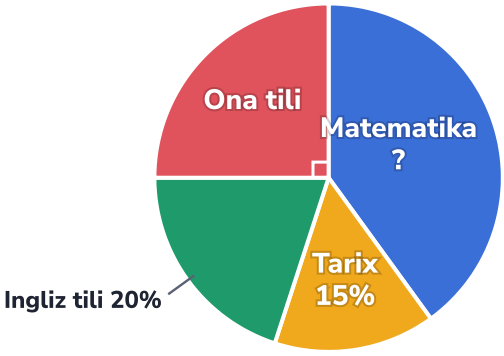
7. Tikuvchilik sexi bir haftada to'rt xil kiyim tikdi. Ko'ylak va shim birga yarim doirani egallaydi.
   a) Ko'ylak va shim birga necha foizni tashkil qiladi?
   b) Jami nechta kiyim tikildi?
   c) Nechta futbolka tikildi?
   d) Ko'ylak va shim soni 1 : 4 nisbatda. Nechta ko'ylak tikildi?



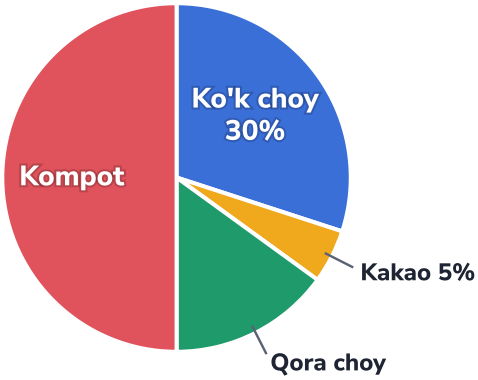
8. Hasharda turli ko'chatlar ekildi. Terak va archa ko'chatlari teng.
   a) Jami nechta ko'chat ekildi?
   b) Ko'chatlarning qancha qismi terak?
   c) Tol va chinor 3 : 1 nisbatda. Nechta chinor ekildi?
   d) Chinorlar sonining archalar soniga nisbati qanday?



9. O'quvchilardan sevimli fani so'raldi. Matematikani 160 o'quvchi tanladi.
   a) Necha foiz o'quvchi ona tilini tanladi?
   b) Necha foizi matematikani tanladi?
   c) So'rovda nechta o'quvchi qatnashdi?
   d) Tarixni nechta o'quvchi tanladi?

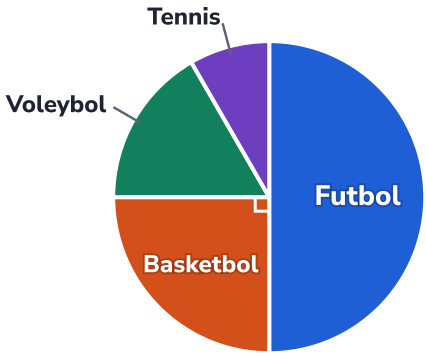


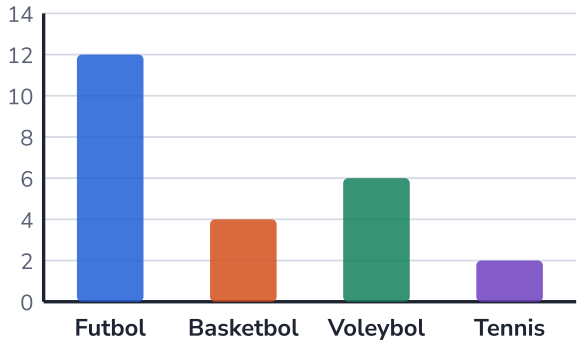
10. Choyxona bir kunda to'rt xil ichimlik sotdi. Ko'k choy va kakao birga 70 ta sotildi. Qora choy kakaodan 3 marta ko'p sotildi.
    a) Jami nechta ichimlik sotildi?
    b) Qora choy nechta sotildi?
    c) Kompot necha foiz?
    d) Qaysi ikki ichimlik birga barcha ichimliklarning $\frac{1}{5}$ qismini tashkil qiladi?



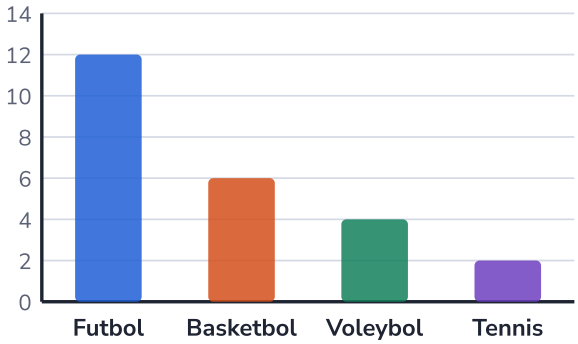
11. Maktab bog'idan 240 kg meva terildi: olma — 80, uzum — 90, anor — 30, nok — 40 kg.
    a) Har bir meva hosilning qancha qismini tashkil qiladi?
    b) Doira 24 ta teng bo'lakka bo'lingan. Har bir meva uchun nechta bo'lakni bo'yash kerak?

12. Sport do'konida bir kunda 24 ta to'p sotildi. Qaysi ustunli diagramma doiraviy diagrammadagi ma'lumotni to'g'ri ko'rsatadi? Tushuntiring.





A



B

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $180 - 64 - 70 = 46$ ta. b) Darslik — yarim doira, u qolgan yarmiga teng: $18 + 27 = 45$ ta. c) $1 - \frac{1}{3} - \frac{1}{8} - \frac{1}{4} = \frac{24}{24} - \frac{8}{24} - \frac{3}{24} - \frac{6}{24} = \frac{7}{24}$. d) Bug'doy — chorak (25%). $100\% - 45\% - 25\% - 16\% = 14\%$.

2-masala

a) 6-sinf — chorak bo'lak: $40 \cdot 4 = 160$ o'quvchi. b) $160 - 42 - 40 - 51 = 27$ o'quvchi.

3-masala

a) Albom — chorak, $\frac{1}{4}$. b) Katakli va chiziqli birga $1 - \frac{1}{4} = \frac{3}{4}$; bu 3 + 1 = 4 ulush, bir ulush $\frac{3}{4} : 4 = \frac{3}{16}$. Katakli — 3 ulush: $\frac{3}{16} \cdot 3 = \frac{9}{16}$. c) $\frac{9}{16}$ qism — 54 ta. $\frac{1}{16}$ qism: $54 : 9 = 6$, jami $6 \cdot 16 = 96$ ta. d) $6 \cdot 3 = 18$ ta.

4-masala

a) $1 - \frac{2}{5} - \frac{1}{4} - \frac{3}{20} = \frac{20}{20} - \frac{8}{20} - \frac{5}{20} - \frac{3}{20} = \frac{4}{20} = \frac{1}{5}$. b) $6 : \frac{3}{20} = 6 \cdot \frac{20}{3} = 40$ soat. c) $\frac{1}{5} \cdot 40 = 8$ soat. d) Basketbol $\frac{2}{5} \cdot 40 = 16$ soat, voleybol $\frac{1}{4} \cdot 40 = 10$ soat. $16 - 10 = 6$ soat.

5-masala

a) Olma — chorak (25%). $100\% - 25\% - 35\% = 40\%$. b) 5 ulush: $40\% : 5 = 8\%$. c) $150 \cdot 4 = 600$ kg. d) Anjir: $600 : 100 \cdot 8 = 48$ kg; $48 \cdot 25\,000 = 1\,200\,000$ so'm.

6-masala

a) O'g'il nemis: $36 - 30 = 6$; qizlar jami: $72 - 36 = 36$; qiz ingliz: $36 - 12 = 24$; ingliz jami: $30 + 24 = 54$; nemis jami: $6 + 12 = 18$. b) $\frac{54}{72} = \frac{3}{4}$, nemis tili — chorak. Javob: B.

7-masala

Yangi jami: $10+6+9=25$. Yashil: $\frac{10}{25}\cdot 100\%=40\%$, sariq: $\frac{6}{25}\cdot 100\%=24\%$, ko'k: $\frac{9}{25}\cdot 100\%=36\%$. Javob: A (B da yashil va ko'k almashib qolgan).

8-masala

a) Birinchi maktabda olma $\frac{20}{40}=50\%$, o'rik $\frac{12}{40}=30\%$, gilos 20%. 100 ta daraxtdan: 50 ta olma, 30 ta o'rik, 20 ta gilos. b) O'zgarmaydi — ikkalasida ham 50%.

9-masala

a) Avtobus — chorak: $360:4=90$ nafar. b) Poyezd va samolyot birga: $360-90-54=216$; 3 ulush, samolyot $216:3=72$ nafar. c) Poyezd: $72\cdot 2=144$ nafar.

10-masala

a) $1-\frac{1}{6}-\frac{1}{8}-\frac{1}{3}=\frac{24}{24}-\frac{4}{24}-\frac{3}{24}-\frac{8}{24}=\frac{9}{24}=\frac{3}{8}$. b) $6:\frac{1}{8}=48$ t; kartoshka $\frac{3}{8}\cdot 48=18$ t. c) To'g'ri emas: sabzi va karam $\frac{4}{24}+\frac{8}{24}=\frac{12}{24}=\frac{1}{2}$, ya'ni 24 t; kartoshka — 18 t.

11-masala

b va d — butun (ovqat, barcha ovozlar) bo'laklarga bo'linadi, doiraviy diagramma qulay. a va c — vaqt bo'yicha o'zgarish, ularni ustunli yoki chiziqli diagrammada ko'rsatgan ma'qul.

Mustaqil ish

1-masala

Dutor — chorak: jami $36\cdot 4=144$. a) Pianino — yarim doira: $144:2=72$ nafar. b) $144-72-36-12=24$ nafar.

2-masala

a) $32:16=2$ marta. b) Chumchuq — chorak, qolgan uchtasi birga $32+12+16=60$ ta — bu $\frac{3}{4}$ qism, $60:3=20$ ta. c) Jami $60+20=80$; $\frac{12}{80}=\frac{3}{20}$.

3-masala

a) $\frac{1}{12} \cdot 3 = \frac{3}{12} = \frac{1}{4}$. b) $1 - \frac{1}{12} - \frac{3}{12} = \frac{8}{12} = \frac{2}{3}$.

4-masala

a) Kunduzgi — yarim doira: $210 \cdot 2 = 420$ ta. b) Kechki — chorak (25%). Tungi: $100\% - 50\% - 25\% - 15\% = 10\%$; $420 : 100 \cdot 10 = 42$ ta.

5-masala

a) Tovuq — 50%, o'rdak — 25%. Kurka: $100\% - 50\% - 25\% - 15\% = 10\%$. b) $50\% : 10\% = 5$ marta. c) $120 \cdot 2 = 240$ ta. d) $240 : 100 \cdot 15 = 36$ ta.

6-masala

a) $\frac{1}{10} \cdot 3 = \frac{3}{10}$. b) She'riyat — chorak: $1 - \frac{1}{10} - \frac{3}{10} - \frac{1}{4} = \frac{20}{20} - \frac{2}{20} - \frac{6}{20} - \frac{5}{20} = \frac{7}{20}$. c) $70 : \frac{7}{20} = 70 \cdot \frac{20}{7} = 200$ ta. d) $\frac{3}{10} \cdot 200 = 60$ ta.

7-masala

a) 50%. b) $120 : \frac{1}{5} = 600$ ta. c) $600 - 300 - 120 = 180$ ta. d) Ko'ylak va shim birga 300 ta, 5 ulush: $300 : 5 = 60$ ta ko'ylak.

8-masala

a) Mevali — yarim doira: $800 \cdot 2 = 1600$ ta. b) $\frac{200}{1600} = \frac{1}{8}$. c) Tol va chinor birga: $1600 - 800 - 200 - 200 = 400$; 4 ulush, chinor $400 : 4 = 100$ ta. d) $100 : 200$, ya'ni $1 : 2$.

9-masala

a) Ona tili — chorak, 25%. b) $100\% - 15\% - 20\% - 25\% = 40\%$. c) 40% — 160, 10% — 40, jami $40 \cdot 10 = 400$ o'quvchi. d) $400 : 100 \cdot 15 = 60$ o'quvchi.

10-masala

a) Ko'k choy va kakao birga $30\% + 5\% = 35\%$ — 70 ta. 5% — 10 ta, jami $10 \cdot 20 = 200$ ta. b) Qora choy — 15% (kakaodan 3 marta ko'p). $200 : 100 \cdot 15 = 30$ ta. c) $100\% - 30\% - 15\% - 5\% = 50\%$. d) $\frac{1}{5} = 20\%$: qora choy va kakao ($15\% + 5\%$).

11-masala

a) Olma $\frac{80}{240} = \frac{1}{3}$, uzum $\frac{90}{240} = \frac{3}{8}$, anor $\frac{30}{240} = \frac{1}{8}$, nok $\frac{40}{240} = \frac{1}{6}$. b) 24 bo'lakdan: olma 8, uzum 9, anor 3, nok 4 ta ($8 + 9 + 3 + 4 = 24$).

12-masala

Futbol — yarim doira: $24 : 2 = 12$. Basketbol — chorak: $24 : 4 = 6$. Qolgan 6 tani voleybol va tennis bo'lishadi, voleybol bo'lagi kattaroq. A da basketbol 4 ta — chorakdan kam. Javob: B.